

Climbing stairs ✓

1 or 2

$n=4 \Rightarrow 5$



$n \rightarrow 1+1+1$
 $1+2$
 $2+1$

$1+1+1+1 \Rightarrow \checkmark$
 $1+1+2 \Rightarrow \checkmark$
 $1+2+1 \Rightarrow \checkmark$
 $2+1+1 \Rightarrow \checkmark$
 $2+2 \Rightarrow \checkmark$

answer

✓

$\Rightarrow 5$

$n=5 \Rightarrow$

$1+1+1+1+1 \checkmark$
 $1+1+1+2 \checkmark$
 $1+1+2+1 \checkmark$
 $1+2+1+1 \checkmark$
 $2+1+1+1 \checkmark$
 $2+2+1 \checkmark$
 $2+1+2 \checkmark$
 $2+2+1 \checkmark$

tarik

6

$n=0$

$n=2$

$1+1$
 2

$n=1$

1

n	Total
1	1
2	2
3	3
4	5

question
 $n=0, 0 \Rightarrow 0$

5 8

4

$$\begin{array}{l}
 1+1+1+1 \quad +1 \\
 1+1+2 \quad +1 \\
 1+2+1 \quad +1 \\
 2+1+1 \quad +1 \\
 2+2 \quad +1
 \end{array}$$

Type

5

climb(n)

1+1+1+1+1

1+2+1+1

Total $\Rightarrow 1+1+1+2$

$\Rightarrow 1+2+2$

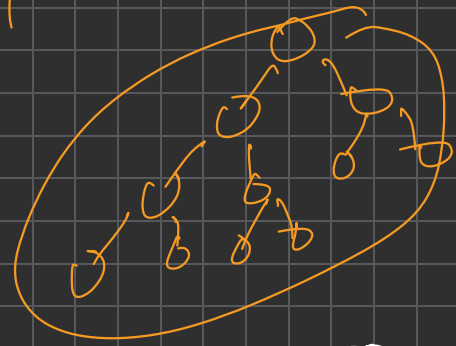
$\Rightarrow 2+1+2$

$climb(5) \Rightarrow climb(4) + climb(3)$

```

int climb (int n)
{
    if (n <= 2)
        return n;

    return climb(n-1) + climb(n-2);
}
    
```

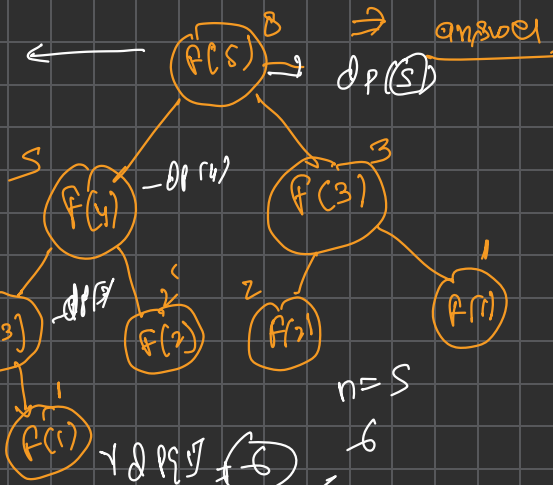


$2^n \Rightarrow$ Recursion

0 - 5 $\Rightarrow$ n+i

dp	0	1	2	3	4	5
	-1	-1	-1	-1	-1	-1

$f(5) \Rightarrow dp(5)$
 $f(4) \Rightarrow dp(4)$
 $f(3) = dp(3)$
 $f(2) = dp(2)$
 $f(1) = dp(1)$



Computat

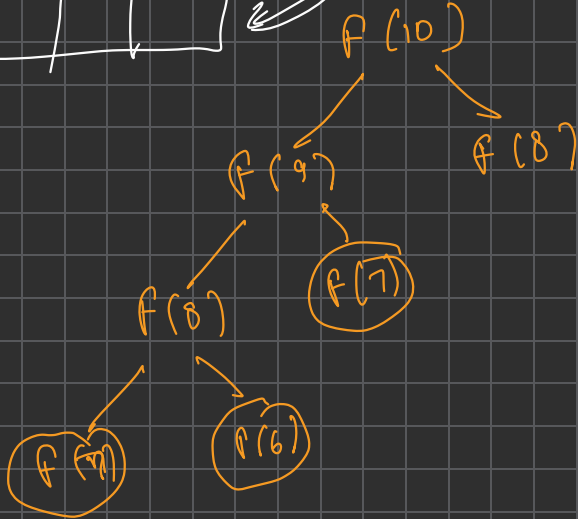
$f(0)$
 $\underline{dp(0)}$

0	1	2	3	4	5

array

$n=5$
6

decomputation

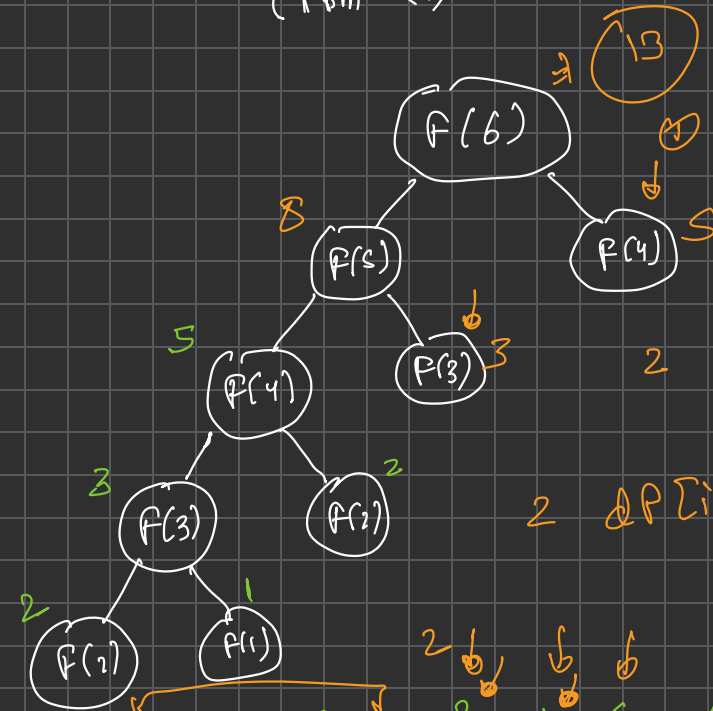


2^n

int climb6(int n)

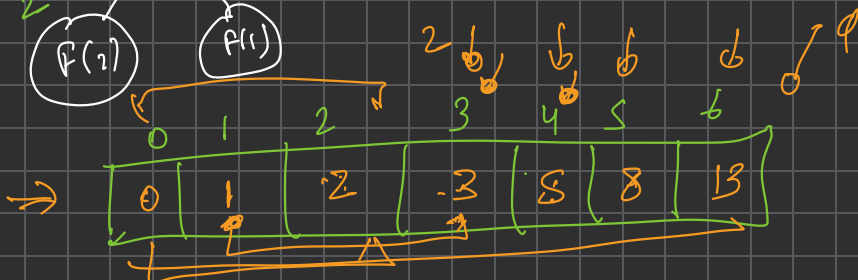
ans = climb6(3)

climb6(3)



$2n$
 $\approx O(n) + O(n)$

$$2 \text{ } dp[i] = dp[i-1] + dp[i-2]$$



$O(n)$
 $O(n)$

→ Storage
 → Optimize

2 | 3 | 5

for (i=3; i<=n; i++)

$dp[0] = dp[1]$

$dp[1] = dp[2]$

$dp[2] = dp[0] + dp[1]$

IO app

IO app & code

1

$O(1)$ at

Time. Com

Space

T. Com

S. Com

T. Com

S. Optimize

Recursive → Top down → Bottom up → Space

↓ Ram constraint ↑

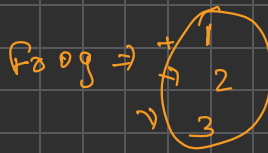
Code run

Space

Ram capacity

Input

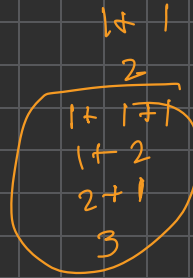
S. Storage



$n=5$

$$Climb(5) = Climb(4) + Climb(3) + Climb(2)$$

$n=0$	1
$n=1$	1
$n=2$	2
$\rightarrow n=3$	4
$\rightarrow n=4$	7
$\rightarrow n=5$	13
$n=6$	24

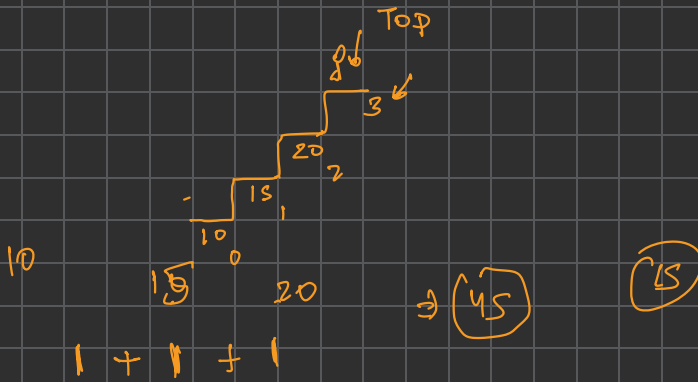


- 1 + 1 + 1 + 1
- 1 + 2 + 1
- 1 + 1 + 2
- 2 + 1 + 1
- 2 + 2
- 1 + 3
- 3 + 1

2	4	7	13
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PO

[10, 15, 20]



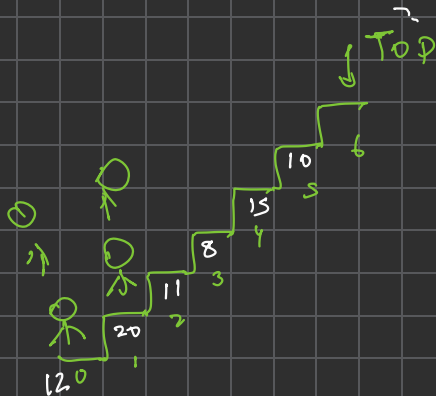
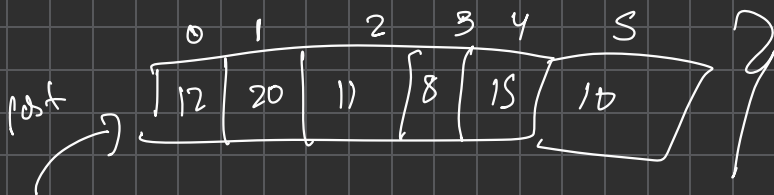
(10, 15, 20) jump

min cost to get

top of

panel

$$15 + 20 = 35$$



$$\min (\text{cost} + \min(\text{cost}[5], 15 + \min(\text{cost}[4])))$$

$$\min(\text{cost}[n]) = \min (\text{cost}[n-1] + \min(\text{cost}[n-1], \text{cost}[n-2] + \min(\text{cost}[n-2])))$$

```
int minCost ( int n )
{
    // Base case
    if ( n <= 1 )
        return 0;
```

$$\min (\text{cost}[n-1] + \min(\text{cost}[n-1], \text{cost}[n-2] + \min(\text{cost}[n-2])))$$

$$\text{Cost}[n-2] + \min(\text{Cost}[n-2]) ;$$

7

